

Chapter Name

Flow of Control

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1. Introduction

2. Types of Statements in Python

What is Statement?

Statements are the instruction given to the computer to perform any kind of action.

3 Types of Statements are:

1. Empty Statement
2. Simple Statement(Single Statement)
3. Compound Statement

1.Empty Statement

The simplest statement is the empty statement,e.e a statement which does nothing.

Keyword of use create empty statement is

In []:

```
pass
```

1. In Python an empty statement is pass statement.
2. The pass statement of Python is a do nothing statement,i.e. empty statement or null operation statement.

2. Simple Statement

Any single executable statement(line) is a simple statement in Python

In []:

For Example:

1. name=input("Enter value of name")
2. print(name)

3. Compound Statement

A compound statement represents a group of statements of Python as a unit.

Format of creating Compound Statement

In []:

```
<compound statement header>:  
    <indented body containing multiple\  
    simple and/or compound statement>  
    <indented body containing multiple\  
    simple and/or compound statement>  
asasdjlkasjdlkasjdlkasjdksaj
```

a Header line:

- Which begins with a keyword and ends with a colon.

a Body:

- Consisting of one or more Python statements(simple Statement).
- Each indented inside the header line.
- All statements in the body are at the same level of indentation.

In [2]:

```
a=int(input("Enter value of a:- "))
b=int(input("Enter value of b:- "))
if a>b:
    print("A greater B")

print("Ok Tata Bye")
```

Enter value of a:- 45

Enter value of b:- 23

A greater B

Ok Tata Bye

3. Statement Flow Control

Types of Excecuton Order of Program:

1. Sequential(Sequence) Exectution
2. Selection(Conditional) Execution
3. Repetition Iteration(Looping)

1.Sequential(Sequence)

1. Sequence - refers to the normal flow of control in a program and is the simplest one.
2. The sequence construct means the statements are being executed sequentially.
3. This represents the default flow of statement.

Selection(Conditional)

- 1.The selection construct means the exectution of statement(s) depending upon a condition-test.
- 2.If a condition evaluates to True - a course of action is followed.
- 3.Otherwise another course of action is followed.
- 4.This construct is also called decision construct

In []:

```
Types of Selection(Conditional) Construct:
1. if
2. if else
3. elif
4. Nested if else
```

The if statements of Python

1. The if statements are the conditional statements in Python and these implement selection constructs(decision constructs).
2. if statement test a particular condition:

if the condition evaluates to true, a course of action performed.

Format of if statement

In []:

```
if condition:
    <if block>
    <if block>
    <if block>
    .
    .
    .
```

WAP to check about EFV or not EFV

In [6]:

```
age = int(input("Enter value of age:- "))
if age>=18:
    print("EFV")

if age<18:
    print("NEFV")
```

Enter value of age:- 4
NEFV

WAP to accept any three subject marks and calculate following things:

1. Sum of all marks
2. average of marks
3. Print "A" if it is more than 50 otherwise print "B"(Average)

In [18]:

```
m1=int(input("Enter marks of subject1:- "))
m2=int(input("Enter marks of subject2:- "))
m3=int(input("Enter marks of subject3:- "))
total = m1+m2+m3
avg = total/3
if avg>=50:
    print("Grade - A","Total:- ",total,"Average:- ",avg)
if avg<50:
    print("Grade - B","Total:- ",total,"Average:- ",avg)
```

Enter marks of subject1:- 10
Enter marks of subject2:- 20
Enter marks of subject3:- 30
Grade - B Total:- 60 Average:- 20.0

WAP to accept any three subject marks and calculate following things:

1. Sum of all marks
2. average of marks
3.
 - a .Print A if marks more than 201 and less than 300
 - b. Print B if marks more than 101 and less than 200
 - c. Print C if marks more than 1 and less than 100

In [3]:

```

m1=float(input("Enter marks of subject1:- "))
m2=float(input("Enter marks of subject2:- "))
m3=float(input("Enter marks of subject3:- "))
total = m1+m2+m3
avg = total/3
if total>=1 and total<=100:
    print("C",total,avg)
if total>=101 and total<=200:
    print("B",total,avg)
if total>=201 and total<=300:
    print("A",total,avg)

```

```

Enter marks of subject1:- 50
Enter marks of subject2:- 50
Enter marks of subject3:- 50
B 150.0 50.0

```

In [25]:

```

m1=float(input("Enter marks of subject1:- "))
m2=float(input("Enter marks of subject2:- "))
m3=float(input("Enter marks of subject3:- "))
total = m1+m2+m3
avg = total/3
if 1<=total<=100:
    print("C",total,avg)
if 101<=total<=200:
    print("B",total,avg)
if 201<=total<=300:
    print("A",total,avg)

```

```

Enter marks of subject1:- 30
Enter marks of subject2:- 40
Enter marks of subject3:- 30.5
C 100.5 33.5
B 100.5 33.5

```

WAP to input day number and display respective day of the week

In [15]:

```
dayn=int(input("Enter day number:- "))
if dayn == 1:
    print("Mon")
if dayn == 2:
    print("Tue")
if dayn == 3:
    print("Wed")
if dayn == 4:
    print("Thu")
if dayn == 5:
    print("Fri")
if dayn == 6:
    print("Sat")
if dayn == 7:
    print("Sun")
if dayn<=0 or dayn>=8:
    print("Wrong Input")
```

Enter day number:- 7
Sun

The if-else Statement

The block below if gets executed if the condition evaluates to true and the block below else gets executed if the condition evaluates to false.

Format of if_else

In []:

```
if condition:
    <if block>
    <if block>
    <if block>
else:
    <else block>
    <else block>
    <else block>
```

WAP to check about EFV or not EFV

In [17]:

```
age = int(input("Enter value of age:- "))
if age>=18:
    print("EFV")
else:
    print("NEFV")
```

Enter value of age:- 5
NEFV

1. Program that takes a number and checks whether the given number is odd or even.

In [28]:

```
no = int(input("Enter value of no:- "))
if no%2 == 0:
    print(no,"is even")
else:
    print(no,"is odd")
```

Enter value of no:- 21
21 is odd

Program to display for calculating area of circle or perimeter of a circle.

(Display AOC if radius is more than 50 otherwise display POC)

In [30]:

```
import math
radius = float(input("Enter Radius Value:- "))
if radius>=50:
    aoc = 3.14 * math.pow(radius,2)    # radius**2
    print("Area of Circle :- ",aoc)
else:
    poc = 2 * 3.14 * radius
    print("Perimeter of Circle :- ",poc)
```

Enter Radius Value:- 20
Perimeter of Circle :- 125.60000000000001

Practice Program(H.W)

1. Program that takes a number and checks whether the given number is odd or even.
2. Program to accept three integers and print the largest of the three. Make use of only if statement.
3. Program that inputs three numbers and calculates two sums as per this:
 - a. sum1 as the sum of all input numbers
 - b. sum2 as the sum of non-duplicate numbers; if there are duplicate numbers in the input, ignores them
4. Program to test the divisibility of a number with another number(i.e., if a number is divisible by another number)
5. Program to find the multiples of a number(the divisor) out of given five numbers.
6. Program to display for calculating area of circle or perimeter of a circle.

In [7]:

```
char = input("Enter a character:- ")
if char=='A':
    print("yes it is A")
else:
    print("it is not A")
```

Enter a character:- f
No it is A

In [9]:

```
a=int(input("Enter value of a:- "))
b=int(input("Enter value of b:- "))
if a>b:
    print("A is greater than b")
else:
    print("B is greater than A")
```

Enter value of a:- 100
Enter value of b:- 20
A is greater than b

In [12]:

```
a=int(input("Enter a value:- "))
if a:
    print("It contains some value Which means condition is True")
else:
    print("It Empty Which means condition is False")
```

Enter a value:- 101-100
It contains some value Which means condition is True

The if - elif statement

Format of if-elif statement

In []:

```
if condition:
    <if block>
elif condition:
    <elif block>
elif condition:
    <elif block>
elif condition:
    <elif block>
elif condition:
    <elif block>
else:
    <else block>          # Optional
```

WAP to accept two numbers and display largest one

In [34]:

```
a=int(input("Enter value of a:- "))
b=int(input("Enter value of b:- "))
if a>b:
    print("A is greater than b")
if b>a:
    print("B is greater than A")
if a==b:
    print("A is equal to B")
```

```
Enter value of a:- 10
Enter value of b:- 10
A is equal to B
```

In [37]:

```
a=int(input("Enter value of a:- "))
b=int(input("Enter value of b:- "))
if a>b:
    print("A is greater than b")
elif b>a:
    print("B is greater than A")
else:
    print("A is equal B")
```

```
Enter value of a:- 10
Enter value of b:- 10
A is equal B
```

In [21]:

```
a=int(input("Enter value of a:- "))
b=int(input("Enter value of b:- "))
if a>b:
    print("A is greater than b")
if b>a:
    print("B is greater than A")
if a==b:
    print("A is equal B")
```

```
Enter value of a:- 10
Enter value of b:- 10
A is equal B
```

Program using multiple if

In [29]:

```
dayn = int(input("Enter day no of week"))
if dayn==1:
    print("Mon")
if dayn==2:
    print("Tue")
if dayn==3:
    print("Wed")
if dayn==4:
    print("Thu")
if dayn==5:
    print("Fri")
if dayn==6:
    print("Sat")
if dayn==7:
    print("Sun")
if dayn>7:
    print("Wrong Day")
```

Enter day no of week4
Thu

Program using elif

In [34]:

```
dayn = int(input("Enter day no of week"))
if dayn==1:
    print("Mon")
elif dayn==2:
    print("Tue")
elif dayn==3:
    print("Wed")
elif dayn==4:
    print("Thu")
elif dayn==5:
    print("Fri")
elif dayn==6:
    print("Sat")
elif dayn==7:
    print("Sun")
else:
    print("Wrong Day")
```

Enter day no of week5
Fri

Prgram that reads two numbers and an-arithmetic operator and display the computed result.

In [38]:

```
n1=int(input("Enter value of n1:- "))
n2=int(input("Enter value of n2:- "))
opt = input("Enter any operator:- ")
if opt=='+':
    result=n1+n2
    print("Result:- ",result)
elif opt=='-':
    result=n1-n2
    print("Result:- ",result)
elif opt=='*':
    result=n1*n2
    print("Result:- ",result)
elif opt=='/':
    result=n1/n2
    print("Result:- ",result)
elif opt=='%':
    result=n1%n2
    print("Result:- ",result)
else:
    print("Wrong Operator")
```

Enter value of n1:- 12
Enter value of n2:- 6
Enter any operator:- %
Result:- 0

Nested if

WAP to accept 3 numbers from user and display largest one

Using Multiple if

In [3]:

```
a=int(input("Enter value of a:- "))
b=int(input("Enter value of b:- "))
c=int(input("Enter value of c:- "))
if a>b and a>c:
    print(a,"is greater than",b,c)
if b>a and b>c:
    print(b,"is greater than",a,c)
if c>a and c>b:
    print(c,"is greater than",a,b)
```

Enter value of a:- 12
Enter value of b:- 45
Enter value of c:- 89
89 is greater than 12 45

In [14]:

```

a=int(input("Enter value of a:- "))
b=int(input("Enter value of b:- "))
c=int(input("Enter value of c:- "))
if a>b:
    if a>c:
        print(a,"greater than",b,c)
    else:
        print(c,"greater than",b,a)
else:
    if b>c:
        print(b,"greater than",a,c)
    else:
        print(c,"greater than",b,a)

```

Enter value of a:- 15
Enter value of b:- -85
Enter value of c:- 45
45 greater than -85 15

WAP to accept 3 number and display in asscending order.

In [15]:

```

a=int(input("Enter value of a:- "))
b=int(input("Enter value of b:- "))
c=int(input("Enter value of c:- "))
#100 50 20
if a!=b and b!=c and a!=c:
    if a>b and a>c:
        if b>c:
            min,mid,max=c,b,a
        else:
            min,mid,max=b,c,a
    elif b>a and b>c:
        if a>c:
            min,mid,max=c,a,b
        else:
            min,mid,max=a,c,b
    else:
        if a>b:
            min,mid,max=b,a,c
        else:
            min,mid,max=a,b,c
    print(min,mid,max)
else:
    print("Checking not possible")

```

Enter value of a:- 10
Enter value of b:- -15
Enter value of c:- 45
-15 10 45

WAP to accept one character and do following things:

1. Check - Uppercase
2. Check - Lowercase
3. Check - Uppercase Vowel
4. Check - Lowercase Vowel
5. Check - 0 to 9

In [7]:

```
chr = input("Enter a character:- ")
if chr>='A' and chr<='Z':
    print("Uppercase")
if chr>='a' and chr<='z':
    print("Lowercase")
if chr=='A' or chr=='E' or chr=='I' or chr=='O' or chr=='U':
    print("Uppercase Vowel")
if chr in ['a','e','i','o','u']:
    print("Lowercase Vowel")
if chr>='0' and chr<='9':
    print("Digit")
```

Enter a character:- r
Lowercase

Storing Condition/Named Condition

1. In Python language we can store any condition's result into identifier which further can be used in program for different purpose
2. You can use named conditions in if statements to enhance readability and reusability of condition.

In []:

```
chr = input("Enter a character:- ")
up = chr>='A' and chr<='Z'
lp = chr>='a' and chr<='z'
upv = chr=='A' or chr=='E' or chr=='I' or chr=='O' or chr=='U'
lpv = chr in ['a','e','i','o','u']
dig = chr>='0' and chr<='9'

if up:
    print("Uppercase")
if lp:
    print("Lowercase")
if upv:
    print("Uppercase Vowel")
if lpv:
    print("Lowercase Vowel")
if dig:
    print("Digit")
```

The range() function

1. The range() function of Python generates a list which is a special sequence type.
2. A sequence in Python is a succession of values bound together by a single name.

Format of range()

In []:

```
range(lowerlimit , upperlimit)
```

In []:

```
range(start,end,step)
range(lowerlimit,upperlimit,stepvalue)
```

In [8]:

```
result = list(range(1,11,1))
print(result)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

In [11]:

```
result = list(range(1,51,5))
print(result)
```

```
[1, 6, 11, 16, 21, 26, 31, 36, 41, 46]
```

In [16]:

```
result = list(range(0,11,0))
print(result)
```

```
-----
-
ValueError                                Traceback (most recent call las
t)
<ipython-input-16-322193279fe2> in <module>()
----> 1 result = list(range(0,11,0))
      2 print(result)
```

ValueError: range() arg 3 must not be zero

In [18]:

```
result = list(range(1,10,-1))
print(result)
```

```
[1, 0, -1, -2, -3, -4, -5, -6, -7, -8, -9]
```

In [23]:

```
result = list(range(11,1,-1))
print(result)
```

```
[11, 10, 9, 8, 7, 6, 5, 4, 3, 2]
```

In [20]:

```
result = list(range(,10,))
print(result)
```

```
File "<ipython-input-20-c9cd8dd0a2f6>", line 1
    result = list(range(,10,))
                        ^
```

SyntaxError: invalid syntax

In [21]:

```
result = list(range(10))
print(result)
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

Iteration/Looping Statements

1. The iteration statements or repetition statements allow a set of instructions to be performed repeatedly until a certain condition is fulfilled.
2. The iteration statements are also called loops or looping statements.
3. Python provides two kinds of loops:
 - a. for loop #entry control loop
 - b. while loop #entry control loop

These two loops also represent by another names:

1. counting loop - the loops that repeat a certain number of times; Python's for loop is a counting loop.
2. conditional loop - the loops that repeat until a certain thing happens,i,e. they keep repeating as long as some condition is true; example while loop

for loop/Counting loop

1. The for loop of Python is designed to process the items of any sequence, such as a list or a string one by one.
2. Each time, when the loop-body is executed is called an iteration.
3. for loop ends when the loop is repeated for the last value of the sequence.
4. The for loop repeats n number of times, where n is the length of sequence given in for loop header's.
5. The loop variable contains the highest value of the list after the for loop is over.
6. You need not pre-defined loop variable of a for-loop

Format of for loop

In []:

```
for <loopvariable> in <sequence>:
    <body of loop>
    <statements_to_repeat>
    <write here>
```

In [6]:

```
for i in [1,2,3,4,5]:
    print("Python",end=" ")
```

Python Python Python Python Python

In [7]:

```
for i in [10,25,385,-784,-855]:
    print("Python",end=" ")
```

Python Python Python Python Python

In [8]:

```
for i in [0,0,0,0,0]:
    print("Python",end=" ")
```

Python Python Python Python Python

In [9]:

```
for i in (0,0,0,0,0):
    print("Python",end=" ")
```

Python Python Python Python Python

In [10]:

```
for i in "moksh":
    print("Python",end=" ")
```

Python Python Python Python Python

WAP to print 1 - 10 numbers

In [11]:

```
for i in [1,2,3,4,5,6,7,8,9,10]:  
    print(i,end=" ")
```

1 2 3 4 5 6 7 8 9 10

WAP to print table of 2

In [12]:

```
for i in [2,4,6,8,10,12,14,16,18,20]:  
    print(i,end=" ")
```

2 4 6 8 10 12 14 16 18 20

In [14]:

```
for i in [1,2,3,4,5,6,7,8,9,10]:  
    print(i,"*",2,"= ",i*2)
```

1 * 2 = 2
2 * 2 = 4
3 * 2 = 6
4 * 2 = 8
5 * 2 = 10
6 * 2 = 12
7 * 2 = 14
8 * 2 = 16
9 * 2 = 18
10 * 2 = 20

WAP to print table of any input number(Taken in runtime)

In [16]:

```
no = int(input("Enter value of no:- "))  
for i in [1,2,3,4,5,6,7,8,9,10]:  
    print(i,"*",no,"= ",i*no)
```

Enter value of no:- 9

1 * 9 = 9
2 * 9 = 18
3 * 9 = 27
4 * 9 = 36
5 * 9 = 45
6 * 9 = 54
7 * 9 = 63
8 * 9 = 72
9 * 9 = 81
10 * 9 = 90

In [19]:

```
for i in [5,6,7,8,9,10,1,2,3,4]:
    print(i,"*",i,"= ",i*i)
```

```
5 * 5 = 25
6 * 6 = 36
7 * 7 = 49
8 * 8 = 64
9 * 9 = 81
10 * 10 = 100
1 * 1 = 1
2 * 2 = 4
3 * 3 = 9
4 * 4 = 16
```

WAP to print 1-50 number

In [22]:

```
for i in range(1,51):
    print(i,end=" ")
```

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28
29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
```

WAP to print following series:

- a. $\frac{1}{2}$ $\frac{2}{4}$ $\frac{3}{6}$ $\frac{4}{8}$
- b. 1 4 7 10 40
- c. 1 -4 7 -10 13 -16 -40

In [8]:

```
n = int(input("Enter number of terms:- "))
for i in range(1,n):
    print(i,"/",i*2,sep="",end=" ")
```

```
Enter number of terms:- 10
1/2 2/4 3/6 4/8 5/10 6/12 7/14 8/16 9/18
```

In [10]:

```
strt = int(input("Enter Start:- "))
end = int(input("Enter End:- "))
if strt<=end:
    for i in range(strt,end):
        print(i,"/",i*2,sep="",end=" ")
else:
    print("No Series Possible")
```

```
Enter Start:- 2
Enter End:- 8
2/4 3/6 4/8 5/10 6/12 7/14
```

Program to print sum of natural numbers between 1 to 7. Print the sum progressively i.e., after adding each natural number, print sum so far.

In [13]:

```
sum=0
for i in range(1,8):
    sum = sum + i

print(sum)
```

28

In [21]:

```
n=int(input("Enter any number"))
if n%2==0:
    print("Even")
else:
    print("Odd")
```

Enter any number12
Even

WAP to print only even numbers present in range of 1 to 20.

In [22]:

```
for i in range(1,21):
    if i%2==0:
        print(i,end=" ")
```

2 4 6 8 10 12 14 16 18 20

In [16]:

```
for i in range(2,21,2):
    print(i,end=" ")
```

2 4 6 8 10 12 14 16 18 20

WAP to count total number of even present in between range of 1 to 50.

In [30]:

```
counte,counto=0,0
for i in range(1,51):
    if i%2==0:
        counte=counte+1
    else:
        counto=counto+1
print("Total Even:- ",counte)
print("Total Odd:- ",counto)
```

Total Even:- 25
Total Odd:- 25

WAP to sum of even and odd present in between range of 1 to 50.

In [1]:

```
sume,sumo = 0,0
for i in range(1,51):
    if i%2==0:
        sume=sume+i
    else:
        sumo=sumo+i
print("Sum of Even:- ",sume)
print("Sum of Odd:- ",sumo)
```

Sum of Even:- 650

Sum of Odd:- 625

In [33]:

```
import math
result = math.factorial(10)
print(result)
```

3628800

WAP to accept a number and calculate/display factorial of same.

In [37]:

```
fact = 1
number=int(input("Enter value of number:- ")) #5
for i in range(1,number+1):
    fact = fact * i

print("Factorial of ",number,"is:= ",fact)
```

Enter value of number:- 10

Factorial of 10 is:= 3628800

2/9 5/13 8/17 11/21 14/25 17/29 20/33 23/37 26/41 29/45

In [4]:

```
N,D=2,9
for i in range(1,11):
    print(N,"/",D,end="  ",sep="")
    N=N+3
    D=D+4
```

2/9
5/13
8/17
11/21
14/25
17/29
20/33
23/37
26/41
29/45

A/B B/C C/D D/E E/F F/G G/H H/I I/J J/K

In [10]:

```
for ascii in range(65,75):
    print(chr(ascii),"/",chr(ascii+1),end="  ",sep="")
```

A/B B/C C/D D/E E/F F/G G/H H/I I/J J/K

In []:

```
a=65
for i in range(1,11):
    print(chr(a),'/',chr(a+1),sep='',end='\t')
    a=a+1
```

1topower2 + 3topower2 + 5topower2 + 7topower2 + 9topower2

In [11]:

```
import math
sum=0
for x in range(1,10,2):
    sum = sum + math.pow(x,2)
print(sum)
```

165.0

WAP to print any two digit number in reverse form

In [26]:

```
a=int(input("Enter any value:- "))
r1 = a//10
r2 = a%10
result = r2*10 + r1
print(result)
```

Enter any value:- 789
168

WAP to print any 3 digit number in reverse form

In [7]:

```
n = int(input("Enter value of n(3 digit only)"))
r1 = n%10
n = n//10
r2 = n%10
r3 = n//10
result = r1 * 100 + r2 * 10 + r3
print(result,sep="")
```

Enter value of n(3 digit only)123
321

The while loop

1. A while loop a conditional loop that will repeat the instructions within itself as long as a conditional remains true(Boolean True or truth value trueval).
2. The variable used in the condition of while loop must have some value before entering into while loop.

Format of while loop

In []:

```
<loop variable>
while <logicalexpression/condition>:
    <loop body>
    <update expression>
```

WAP to print series for 1 to 10 numbers.

In [10]:

```
i=1
while i<=10:
    print(i)
    i=i+1
print("Bye Bye")
```

```
1
2
3
4
5
6
7
8
9
10
Bye Bye
```

WAP to print table of any number

In [15]:

```
n=int(input("Enter value of n:- "))
i=1
while i<=10:
    print(i*n,end=" ")
    i=i+1
```

```
Enter value of n:- 45
45 90 135 180 225 270 315 360 405 450
```

WAP to print only those numbers where unit digit is 5 in range of 1 to 100.

In [17]:

```
n=1
while n<=100:
    if n%10==5:
        print(n,end=" ")
    n+=1
```

```
5 15 25 35 45 55 65 75 85 95
```

WAP to display factorial of input number.

In [18]:

```

num = int(input("enter a number: "))
fact = 1
i = 1
while i <= num:
    fact = fact * i
    i = i + 1
print("factorial of ", num, " is ", fact)

```

```

enter a number: 5
factorial of 5 is 120

```

In [20]:

```

num = int(input("Enter a number: "))
factorial = 1
i=1
if num < 0:
    print(" Factorial does not exist for negative numbers")
elif num == 0:
    print("The factorial of 0 is 1")
else:
    while i <= num:
        factorial = factorial * i
        i = i + 1
    print("The factorial of",num,"is",factorial)

```

```

Enter a number: 5
The factorial of 5 is 120

```

In [21]:

```

num = int(input("enter a number: "))
fact = 1
while num:
    fact = fact * num
    num=num-1
print("factorial of is:- ", fact)

```

```

enter a number: 5
factorial of is:- 120

```

Write a Python program to print every integer between 1 and n divisible by m. Also report whether the number that is divisible by m is even or odd.

Write a program to display 10 terms of fibonacci series

In []:

```

0 1 1 2 3 5 8 13 21 34

```

In [6]:

```
term = int(input("Enter number of term:- "))
f,s = 0,1
print(f,s,end=" ")
for i in range(1,term-1):
    t = f + s
    print(t,end=" ")
    f = s
    s = t
```

Enter number of term:- 10
 0 1 1 2 3 5 8 13 21 34

WAP to check whehter input number is prime number or not prime number.

In [21]:

```
num = int(input("Enter value of num:- "))
i = 1
count=0
while i<=num:
    if num%i == 0:
        count=count+1
    i=i+1

if count==2:
    print("Prime")
else:
    print("Not Prime")
```

Enter value of num:- 5
 Prime

Loop else Statement

The else clause of a Python loop executes when the loop terminates normally,not when the loop is terminating because of a break statement.

Format of loop else statement

In []:

```
for <loopvariable> in <sequence>:
    <loop body>
    .
    .
else:
    <else block>
```

In [2]:

```
for i in range(1,11):
    if i==5:
        break
    print(i,end=" ")
else:
    print("I successfully complete loop cycle")
```

1 2 3 4

In [5]:

```
i=1
while i<=10:
    if i==5:
        break
    print(i,end=" ")
    i+=1
else:
    print("I successfully complete loop cycle")
```

1 2 3 4

Program to find input is Prime or not Prime

In [14]:

```
num = int(input("Enter value of num:- ")) #7
for i in range(2,num):
    if num%i == 0:
        print("It is Not Prime Number")
        break
else:
    print("It is Prime Number")
```

Enter value of num:- 563

It is Prime Number

Jump Statements - break and continue

break

1. A break statement skips the rest of the loop and jumps over to the statement following the loop.
2. break can not be used outside loop.

In [1]:

```

a=b=c=0
for i in range(1,4):
    a=int(input("Enter the value of a:- "))
    b=int(input("Enter the value of b:- "))
    if b==0:
        break
        print("b is zero")
    else:
        c=a/b
        print(a,"/",b,"= ",c)
print("Bye Bye Bye")

```

```

Enter the value of a:- 10
Enter the value of b:- 0
b is zero
Enter the value of a:- 10
Enter the value of b:- 2
10 / 2 = 5.0
Enter the value of a:- 20
Enter the value of b:- 0
b is zero

```

In [3]:

```

a=b=c=0
for i in range(1,4):
    a=int(input("Enter the value of a:- "))
    b=int(input("Enter the value of b:- "))
    break
    c=a/b
    print(a,"/",b,"= ",c)

```

```

Enter the value of a:- 10
Enter the value of b:- 2

```

Infinite Loops and break statement

In [8]:

```

i=1
while True:
    print(i,end=" ")
    if i==100:
        break
    i=i+1

```

1

Program to implement 'guess the number' game. Python generates a number randomly in the range[10,50]. The user is given five chances to guess a number in the range $10 \leq \text{number} \leq 50$.

1. If the user's guess matches the number generated , Python displays 'You Win' and the loop terminates without completing the five iterations.
2. If, however , cannot guess the number in five attempts, Python displays 'You lose'.

The continue Statement

1. The continue statement skips the rest of the loop statements and causes the next iteration of the loop to take place.

In [1]:

```
a=b=c=0
for i in range(1,4):
    a=int(input("Enter the value of a:- "))
    b=int(input("Enter the value of b:- "))
    if b==0:
        continue
    print("b is zero")
    else:
        c=a/b
        print(a,"/",b,"= ",c)
```

```
Enter the value of a:- 10
Enter the value of b:- 0
Enter the value of a:- 10
Enter the value of b:- 5
10 / 5 = 2.0
Enter the value of a:- 20
Enter the value of b:- 0
```

Nested Loop

A loop may contain another loop in its body, this form of a loop is called nested loop.

Format of nested loop

In []:

```
for <loopvariable> in <sequence>:
    <outerloop body>
    .
    .
    for <loopvariable> in <sequence>:
        <innerloop body>
        .
        .
```

Program to print 5 stars in one line

In [21]:

```
for r in range(1,6):      #OuterLoop controls number of rows
    for i in range(1,6):  #InnerLoop controls number of column(stars)
        print("*",end=" ")
    print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

In []:

```
*
* *
* * *
* * * *
* * * * *
```

In [27]:

```
for r in range(1,6):      #OuterLoop controls number of rows
    for i in range(1,r+1): #InnerLoop controls number of column(stars)
        print("*",end=" ")
    print()
```

```
*
* *
* * *
* * * *
* * * * *
```

In []:

```
* * * * *
* * * *
* * *
* *
*
```

In [7]:

```
for i in range(0,5):
    for j in range(1,6-i):
        print("*",end=" ")
    print()
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

In []:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

In [5]:

```
for r in range(1,6):      #Outerloop controls number of rows
    for i in range(1,r+1): #Innerloop controls number of column(stars)
        print(i,end=" ")
    print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

In []:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

In [8]:

```
for i in range(0,5):
    for j in range(1,6-i):
        print(j,end=" ")
    print()
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

In []:

```
5
5 4
5 4 3
5 4 3 2
5 4 3 2 1
```

In [29]:

```

for r in range(5,0,-1):      #Outerloop controls number of rows
    for i in range(5,r-1,-1): #Innerloop controls number of column(stars)
        print(i,end=" ")
    print()

```

```

5
5 4
5 4 3
5 4 3 2
5 4 3 2 1

```

In [34]:

```

for i in range(1,6):
    n=5
    for j in range(1,i+1):
        print(n,end=" ")
        n=n-1
    print()

```

```

5
5 4
5 4 3
5 4 3 2
5 4 3 2 1

```

In []:

```

5 4 3 2 1
5 4 3 2
5 4 3
5 4
5

```

In [36]:

```

for i in range(0,5):
    n=5
    for j in range(1,6-i):
        print(n,end=" ")
        n=n-1
    print()

```

```

5 4 3 2 1
5 4 3 2
5 4 3
5 4
5

```

In []:

```

A
A B
A B C
A B C D
A B C D E

```

In []:

```
A B C D E
A B C D
A B C
A B
A
```

29.09.21

Output/Error & Practice Program based Questions

Q.1

In [4]:

```
if (0,):
    print('foo')
else:
    print('too')
```

foo

Q.2

In [9]:

```
if (1,2)
    print('foo')
else:
    print('koo')
```

```
File "<ipython-input-9-6790b398f510>", line 1
    if (1,2)
        ^
```

SyntaxError: invalid syntax

Q.3

In []:

```
if (1,2):
    print('foo')
```

Q.4

In [10]:

```
if (1):  
    print('foo')  
else:  
    print('koo')
```

foo

Q.5

In [12]:

```
x=3  
if x==0:  
    print("Am I here?",end=' ')  
elif x==3:  
    print("Or here?",end=' ')  
else:  
    pass  
print("Or Over here?")
```

Or here? Or Over here?

Q.6

In [13]:

```
x = float(input())  
if (x==1):  
    print("yes")  
elif (x>=2):  
    print("Maybe")  
else:  
    print("No")
```

2
Maybe

Q.7

In [18]:

```
a=int(input())
b=int(input())
if a<=0:
    b=b+1
else:
    a=a+1
if a>0 and b>0:
    print("W")
elif a>0:
    print("X")
if b>0:
    print("Y")
else:
    print("Z")
```

-1
-1
Z

In [19]:

```
# 8th questions
a=0
for i in range(4,8):
    if i%2==0:
        a=a+i
print(a)
```

10

In [22]:

```
#9th Program
for i in range(-5):
    print(i)
```

In [23]:

```
#10 Program
for i in range(10,5,-3):
    print(i)
```

10
7

In [26]:

```
#11 Program
for i in range(3):
    pass
```

In [27]:

```
#12 Program
if (4+5 == 10):
    print("TRUE")
else:
    print("FALSE")
print("TRUE")
```

FALSE

TRUE

In [28]:

```
#13 Program
x=1
if x>3:
    if x>4:
        print("A",end=" ")
    else:
        print("B",end=" ")
elif x<2:
    if (x!=0):
        print("C",end=" ")
print("D")
```

C D

In [30]:

```
if int('zero') == 0:
    print('zero')
elif str(0) == 'zero':
    print(0)
elif str(0) == '0':
    print(str(0))
else:
    print("none of the above")
```

-

ValueError Traceback (most recent call last)
t)

<ipython-input-30-4e2c4c57f6f4> in <module>()

```
----> 1 if int('zero') == 0:
      2     print('zero')
      3 elif str(0) == 'zero':
      4     print(0)
      5 elif str(0) == '0':
```

ValueError: invalid literal for int() with base 10: 'zero'

In [31]:

```
n=int(input())
if n<1:
    print("invalid value")
else:
    for i in range(1,n+1):
        print(i*i)
```

3
1
4
9

In [32]:

```
n=int(input())
if n>0:
    for a in range(1,n+1):
        print(a/(n//2))
else:
    print("Now Quitting")
```

3
1.0
2.0
3.0
4.0
5.0

In [33]:

```
for x in [1,2,3,4,5]:
    print(x)
```

1
2
3
4
5

In [34]:

```
for x in range(5):
    print(x)
```

0
1
2
3
4

In [35]:

```
for p in range(1,10):  
    print(p)
```

1
2
3
4
5
6
7
8
9

In [36]:

```
for q in range(100,50,-10):  
    print(q)
```

100
90
80
70
60

In [37]:

```
for z in range(-500,500,100):  
    print(z)
```

-500
-400
-300
-200
-100
0
100
200
300
400

In [39]:

```
x=10  
y=5  
for i in range(x-y*2):  
    print("%",i)
```

In [40]:

```
for x in [1,2,3]:
    for y in [4,5,6]:
        print(x,y)
```

```
1 4
1 5
1 6
2 4
2 5
2 6
3 4
3 5
3 6
```

In [41]:

```
for x in range(3):
    for y in range(4):
        print(x,y,x+y)
```

```
0 0 0
0 1 1
0 2 2
0 3 3
1 0 1
1 1 2
1 2 3
1 3 4
2 0 2
2 1 3
2 2 4
2 3 5
```

In [42]:

```
c=0
for x in range(10):
    for y in range(5):
        c+=1
print(c)
```

50

In [43]:

```
for i in range(4):
    for j in range(5):
        if i+1 == j or j+i == 4:
            print("+",end = ' ')
        else:
            print("o",end=' ')
print()
```

```
o + o o + o o + + o o o + + o o + o o +
```

End of Chapter

Next Chapter - String Manipulation